

Microservices

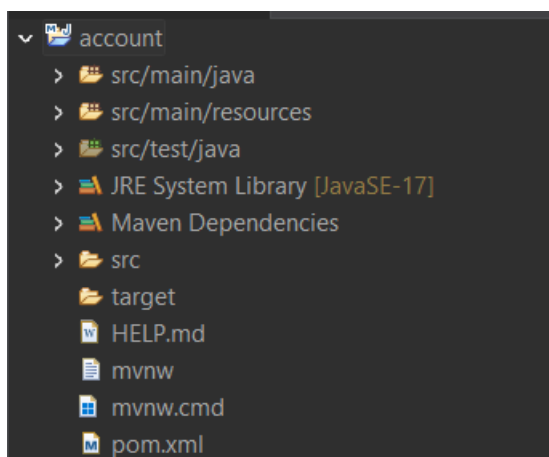
Creating Microservices for account and loan

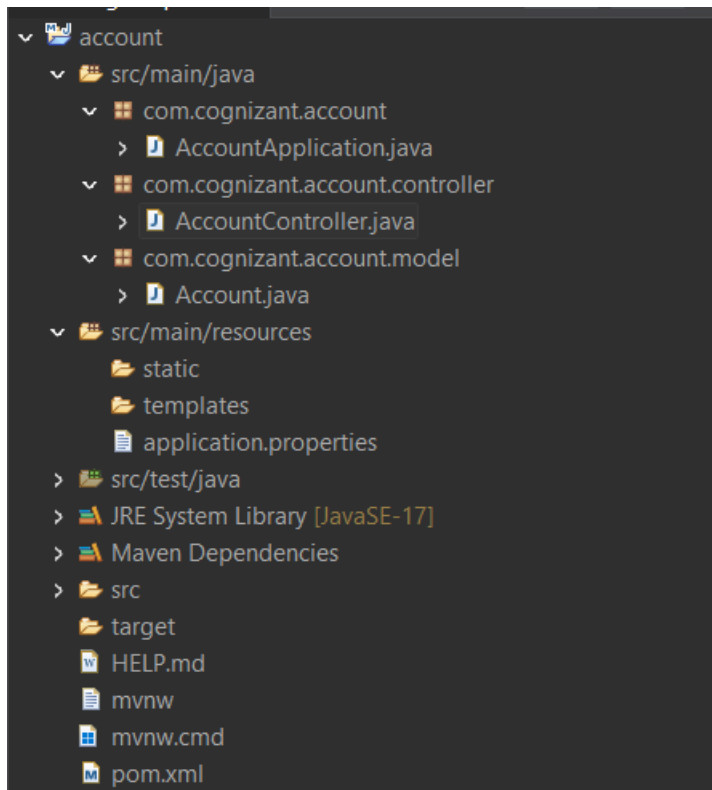
I) Creating account

The Spring Initializr web interface is shown with a dark theme. The left sidebar contains a hamburger menu and a refresh icon. The main content area is divided into several sections:

- Project:** Radio buttons for `Gradle - Groovy`, `Gradle - Kotlin`, and `Maven` (selected).
- Language:** Radio buttons for `Java` (selected), `Kotlin`, and `Groovy`.
- Spring Boot:** Radio buttons for `4.1.1 (SNAPSHOT)`, `4.1.0`, `4.0.8 (SNAPSHOT)`, `4.0.7`, and `3.5.16` (selected).
- Project Metadata:** Text input fields for `Group` (com.cognizant), `Artifact` (account), and `Package name` (com.cognizant.account).
- Packaging:** Radio buttons for `Jar` (selected) and `War`.
- Configuration:** Radio buttons for `Properties` (selected) and `YAML`.
- Java:** Radio buttons for `26`, `25`, `21`, and `17` (selected).
- Dependencies:** A section with a button `ADD DEPENDENCIES... CTRL + B`. It lists `Spring Boot DevTools` (DEVELOPER TOOLS) and `Spring Web` (WEB).

At the bottom, there are three buttons: `GENERATE CTRL + G`, `EXPLORE CTRL + SPACE`, and an ellipsis button.





```
1 package com.cognizant.account.controller;
2
3 import org.springframework.web.bind.annotation.GetMapping;
4 import org.springframework.web.bind.annotation.PathVariable;
5 import org.springframework.web.bind.annotation.RestController;
6
7 import com.cognizant.account.model.Account;
8
9 @RestController
10 public class AccountController {
11
12     @GetMapping("/accounts/{number}")
13     public Account getAccount(@PathVariable String number) {
14
15         Account account = new Account(
16             number,
17             "Savings",
18             234343
19         );
20
21         return account;
22     }
23 }
```

```
1 package com.cognizant.account.model;
2
3 public class Account {
4
5     private String number;
6     private String type;
7     private double balance;
8
9     public Account() {
10    }
11
12    public Account(String number, String type, double balance) {
13        this.number = number;
14        this.type = type;
15        this.balance = balance;
16    }
17
18    public String getNumber() {
19        return number;
20    }
21
22    public void setNumber(String number) {
23        this.number = number;
24    }
25
26    public String getType() {
27        return type;
28    }
29
30    public void setType(String type) {
31        this.type = type;
32    }
33
34    public double getBalance() {
35        return balance;
36    }
```

```
37
38    public void setBalance(double balance) {
39        this.balance = balance;
40    }
41 }
42
```

```

1 package com.cognizant.account;
2
3 import org.springframework.boot.SpringApplication;
4
5
6 @SpringBootApplication
7 public class AccountApplication {
8
9     public static void main(String[] args) {
10         SpringApplication.run(AccountApplication.class, args);
11     }
12
13 }
14

```

```

7 public class AccountApplication {

```

Console x

AccountApplication [Java Application] C:\Program Files\Java\jdk-17\bin\javaw.exe (7 Jul 2026, 7:44:57 pm) [pid: 7912]



:: Spring Boot :: (v3.5.16)

```

2026-07-07T19:44:58.835+05:30 INFO 7912 --- [account] [ restartedMain] c.cognizant.account.AccountApplica
2026-07-07T19:44:58.840+05:30 INFO 7912 --- [account] [ restartedMain] c.cognizant.account.AccountApplica
2026-07-07T19:44:58.943+05:30 INFO 7912 --- [account] [ restartedMain] .e.DevToolsPropertyDefaultsPostPro
2026-07-07T19:44:58.944+05:30 INFO 7912 --- [account] [ restartedMain] .e.DevToolsPropertyDefaultsPostPro
2026-07-07T19:45:00.162+05:30 INFO 7912 --- [account] [ restartedMain] o.s.b.w.embedded.tomcat.TomcatWebS
2026-07-07T19:45:00.182+05:30 INFO 7912 --- [account] [ restartedMain] o.apache.catalina.core.StandardSer
2026-07-07T19:45:00.182+05:30 INFO 7912 --- [account] [ restartedMain] o.apache.catalina.core.StandardEng
2026-07-07T19:45:00.218+05:30 INFO 7912 --- [account] [ restartedMain] o.a.c.c.C.[Tomcat].[localhost].[/]
2026-07-07T19:45:00.220+05:30 INFO 7912 --- [account] [ restartedMain] w.s.c.ServletWebServerApplicationC
2026-07-07T19:45:00.669+05:30 INFO 7912 --- [account] [ restartedMain] o.s.b.d.a.OptionalLiveReloadServer
2026-07-07T19:45:00.709+05:30 INFO 7912 --- [account] [ restartedMain] o.s.b.w.embedded.tomcat.TomcatWebS
2026-07-07T19:45:00.722+05:30 INFO 7912 --- [account] [ restartedMain] c.cognizant.account.AccountApplica

```

```
pplication : Starting AccountApplication using Java 17.0.12 with PID 7912 (C:\Users\deepi\Downl
pplication : No active profile set, falling back to 1 default profile: "default"
ostProcessor : Devtools property defaults active! Set 'spring.devtools.add-properties' to 'false'
ostProcessor : For additional web related logging consider setting the 'logging.level.web' proper
atWebServer : Tomcat initialized with port 9090 (http)
ardService : Starting service [Tomcat]
ardEngine : Starting Servlet engine: [Apache Tomcat/10.1.55]
t].[/] : Initializing Spring embedded WebApplicationContext
ationContext : Root WebApplicationContext: initialization completed in 1272 ms
Server : LiveReload server is running on port 35729
atWebServer : Tomcat started on port 9090 (http) with context path '/'
pplication : Started AccountApplication in 2.435 seconds (process running for 2.853)
```

http://localhost:9090/accounts/12345

GET http://localhost:9090/accounts/12345

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Query Params

Key	Value
Key	Value

Body Cookies Headers 5 Test Results

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```
1 {
2   "number": "12345",
3   "type": "Savings",
4   "balance": 234343.0
5 }
```

II) Creating Loan

The Spring Initializr web interface is shown in a dark theme. It features a sidebar with a hamburger menu and a refresh icon. The main content area is divided into sections for Project, Language, Spring Boot, Project Metadata, and Dependencies. The Project section has radio buttons for Gradle - Groovy, Gradle - Kotlin, and Maven (selected). The Language section has radio buttons for Java (selected), Kotlin, and Groovy. The Spring Boot section has radio buttons for 4.1.1 (SNAPSHOT), 4.1.0, 4.0.8 (SNAPSHOT), 4.0.7, and 3.5.16 (selected). The Project Metadata section has input fields for Group (com.cognizant), Artifact (loan), and Package name (com.cognizant.loan). The Packaging section has radio buttons for Jar (selected) and War. The Configuration section has radio buttons for Properties (selected) and YAML. The Java section has radio buttons for 26, 25, 21, and 17 (selected). The Dependencies section has a button to add dependencies and lists Spring Boot DevTools and Spring Web. At the bottom, there are buttons for GENERATE (CTRL + G), EXPLORE (CTRL + SPACE), and a menu icon.

Project

☐ Gradle - Groovy ☒ Gradle - Kotlin ☒ Maven

Language

☒ Java ☐ Kotlin ☐ Groovy

Spring Boot

☐ 4.1.1 (SNAPSHOT) ☐ 4.1.0 ☐ 4.0.8 (SNAPSHOT) ☐ 4.0.7 ☒ 3.5.16

Project Metadata

Group

Artifact

Package name

Packaging ☒ Jar ☐ War

Configuration ☒ Properties ☐ YAML

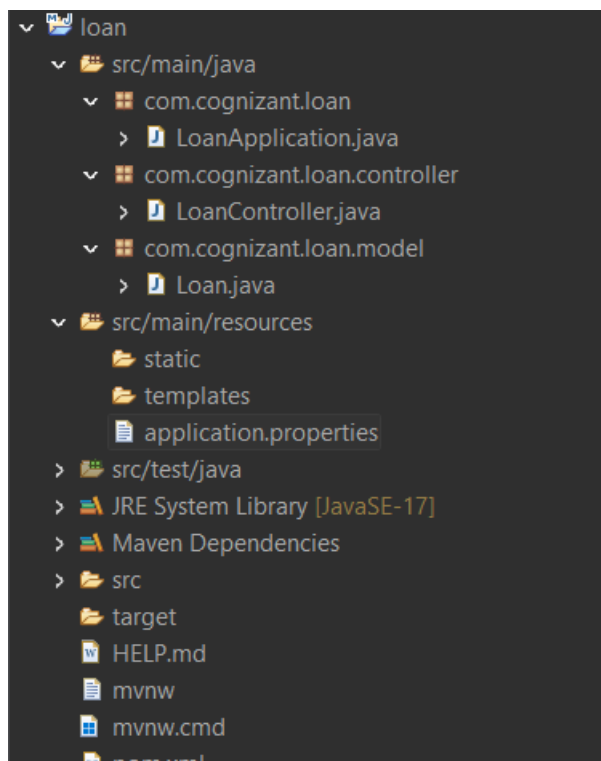
Java ☐ 26 ☐ 25 ☐ 21 ☒ 17

Dependencies [ADD DEPENDENCIES... CTRL + B](#)

Spring Boot DevTools **DEVELOPER TOOLS**
Provides fast application restarts, LiveReload, and configurations for enhanced development experience.

Spring Web **WEB**
Build web, including RESTful, applications using Spring MVC. Uses Apache Tomcat as the default embedded container.

GENERATE CTRL + G **EXPLORE** CTRL + SPACE ...



```

1 package com.cognizant.loan;
2
3 import org.springframework.boot.SpringApplication;
4
5
6 @SpringBootApplication
7 public class LoanApplication {
8
9     public static void main(String[] args) {
10         SpringApplication.run(LoanApplication.class, args);
11     }
12
13 }
14

```

```

1 package com.cognizant.loan.controller;
2
3 import org.springframework.web.bind.annotation.GetMapping;
4 import org.springframework.web.bind.annotation.PathVariable;
5 import org.springframework.web.bind.annotation.RestController;
6
7 import com.cognizant.loan.model.Loan;
8
9 @RestController
10 public class LoanController {
11
12     @GetMapping("/loans/{number}")
13     public Loan getLoan(@PathVariable String number) {
14
15         Loan loan = new Loan(
16             number,
17             "Car",
18             400000,
19             3258,
20             18
21         );
22
23         return loan;
24     }
25
26 }

```

```
1 package com.cognizant.loan.model;
2
3 public class Loan {
4
5     private String number;
6     private String type;
7     private double loan;
8     private int emi;
9     private int tenure;
10
11     public Loan() {
12     }
13
14     public Loan(String number, String type, double loan, int emi, int tenure) {
15         this.number = number;
16         this.type = type;
17         this.loan = loan;
18         this.emi = emi;
19         this.tenure = tenure;
20     }
21
22     public String getNumber() {
23         return number;
24     }
25
26     public void setNumber(String number) {
27         this.number = number;
28     }
29
30     public String getType() {
31         return type;
32     }
33
34     public void setType(String type) {
35         this.type = type;
36     }
37 }
```



```
37
38     public double getLoan() {
39         return loan;
40     }
41
42     public void setLoan(double loan) {
43         this.loan = loan;
44     }
45
46     public int getEmi() {
47         return emi;
48     }
49
50     public void setEmi(int emi) {
51         this.emi = emi;
52     }
53
54     public int getTenure() {
55         return tenure;
56     }
57
58     public void setTenure(int tenure) {
59         this.tenure = tenure;
60     }
61 }
```

```
1spring.application.name=loan
2server.port = 9091
```


Both Loan and account are running simultaneously as they are running in different ports. Account in port 9090 and Loan in port 9091.

```
AccountApplication [Java Application] C:\Program Files\Java\jdk-17\bin\javaw.exe (7 Jul 2026, 8:17:04 pm) [pid: 15600]

:: Spring Boot :: (v3.5.16)

2026-07-07T20:17:05.619+05:30 INFO 15600 --- [account] [ restartedMain] c.cognizant.account.AccountApplic
2026-07-07T20:17:05.624+05:30 INFO 15600 --- [account] [ restartedMain] c.cognizant.account.AccountApplic
2026-07-07T20:17:05.734+05:30 INFO 15600 --- [account] [ restartedMain] .e.DevToolsPropertyDefaultsPostPro
2026-07-07T20:17:05.735+05:30 INFO 15600 --- [account] [ restartedMain] .e.DevToolsPropertyDefaultsPostPro
2026-07-07T20:17:07.485+05:30 INFO 15600 --- [account] [ restartedMain] o.s.b.w.embedded.tomcat.TomcatWeb
2026-07-07T20:17:07.502+05:30 INFO 15600 --- [account] [ restartedMain] o.apache.catalina.core.StandardSe
2026-07-07T20:17:07.502+05:30 INFO 15600 --- [account] [ restartedMain] o.apache.catalina.core.StandardEn
2026-07-07T20:17:07.560+05:30 INFO 15600 --- [account] [ restartedMain] o.a.c.c.C.[Tomcat].[localhost].[/
2026-07-07T20:17:07.564+05:30 INFO 15600 --- [account] [ restartedMain] w.s.c.ServletWebServerApplication
2026-07-07T20:17:08.195+05:30 WARN 15600 --- [account] [ restartedMain] o.s.b.d.a.OptionalLiveReloadServe
2026-07-07T20:17:08.255+05:30 INFO 15600 --- [account] [ restartedMain] o.s.b.w.embedded.tomcat.TomcatWeb
2026-07-07T20:17:08.272+05:30 INFO 15600 --- [account] [ restartedMain] c.cognizant.account.AccountApplic
```

```
: Starting AccountApplication using Java 17.0.12 with PID 15600 (C:\Users\deepi\Downloads\A
: No active profile set, falling back to 1 default profile: "default"
: Devtools property defaults active! Set 'spring.devtools.add-properties' to 'false' to dis
: For additional web related logging consider setting the 'logging.level.web' property to '
: Tomcat initialized with port 9090 (http)
: Starting service [Tomcat]
: Starting Servlet engine: [Apache Tomcat/10.1.55]
: Initializing Spring embedded WebApplicationContext
: Root WebApplicationContext: initialization completed in 1826 ms
: Unable to start LiveReload server
: Tomcat started on port 9090 (http) with context path '/'
: Started AccountApplication in 3.349 seconds (process running for 3.864)
```

GET

▼

http://localhost:9091/loans/H12345

☰ Docs

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Key
Key

Body

Cookies

Headers 5

Test Results

🕒

{ } JSON ▼

▶ Preview

🖼️ Visualize

▼

```
1 {
2   "number": "H12345",
3   "type": "Car",
4   "loan": 400000.0,
5   "emi": 3258,
6   "tenure": 18
7 }
```

GET

⌵

http://localhost:9090/accounts/12345

☰ Docs

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Test Results

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```
1 {
2   "number": "12345",
3   "type": "Savings",
4   "balance": 234343.0
5 }
```